

CISC7021 Applied NLP — Assignment 1: Build a Toy Llama-2 Language Model: Decoding, Perplexity, and Continual Pre-training on Chinese

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Abstract

This report studies a 42M-parameter toy Llama-2 pre-trained on TinyStories. I first generate English stories from three prompts under six decoding settings and compare outputs with lexical statistics. Perplexity (PPL) on English, Chinese and Portuguese test sets shows the model is fluent in English (PPL 4.14) but far out of distribution for Chinese ($\approx 70,004$) and Portuguese ($\approx 25,047$). I then continually pre-train on 10,000 machine-translated Chinese stories in four runs that vary learning rate, epochs, and data. Every run cuts Chinese PPL by orders of magnitude (e.g. $70,004 \rightarrow 3.47$), but English forgetting varies widely; adding 2,000 new stories (Run D) gives the best balance (zh 9.33, en 5.74).

1 Introduction

This assignment studies a 42M-parameter toy decoder-only transformer (Llama-2 architecture) pre-trained on the English TinyStories corpus (1; 2). T1 (decoding) compares six sampling settings on three English prompts; T2 scores the base model on English, Chinese and Portuguese; T3 fine-tunes it on Chinese and tunes hyper-parameters; T4 (optional) grows the Chinese corpus. All numbers come from logs in `assignment1_artifacts/` and `saved_models/`.

2 Experimental Setup

2.1 Hardware and model

All experiments ran on a laptop with an NVIDIA RTX 4070 Laptop GPU (8 GB), PyTorch in mixed precision (fp16). The provided `llama-42m` checkpoint (Llama decoder + tokenizer, loaded with `LlamaForCausalLM`) has 41.69M trainable parameters excluding embeddings (58.07M total; input and output embeddings are tied). Seed is fixed to 42 for every run.

2.2 Data and preprocessing

The Chinese corpus is machine-translated children’s stories. For Task 4 I collected 2,000 more TinyStories, translated them with NLLB-200 (distilled 600M) (3), and concatenated with the 10,000 training docs. QC (`task4_qc.json`) found no empty, too-short or U+FFFD-contaminated items; average length 237.6 chars vs. 258.4 for the original. Text is tokenized with the Llama tokenizer and packed into 512-token blocks (`labels == input_ids`). PPL is the exponentiated mean per-token cross-entropy on 1,000-document test slices. Table 1 summarises the corpus.

Table 1: Corpus statistics.

Split	Docs	512-tok blocks
zh_train (T3)	10,000	7,960
zh_dev	500	393
zh/en/pt_test (each)	1,000	–
T4 merged train	12,000	9,383

2.3 Continual pre-training runs

HuggingFace `Trainer`: linear LR schedule, 1% warm-up, weight decay 0.01, AdamW, batch 32, block 512, fp16. Validated on `zh_dev` every 200 steps (100 for Run D). Table 2 lists the four runs. Runs A–C share the 10,000-doc corpus; Run D uses 12,000. Runs A and B both ran 1,992 steps, but A’s wall-clock time was about twice B’s (~ 179 vs. ~ 84 min; `train_A.json/train_B.json`), so the two times are not directly comparable.

Table 2: Continual pre-training runs and wall-clock time.

Run	LR	Ep	Train docs	Steps (time)
A	10^{-4}	8	10,000	1,992 (~ 179 min)
B	5×10^{-5}	8	10,000	1,992 (~ 84 min)
C	10^{-4}	2	10,000	498 (~ 55 min)
D	10^{-4}	2	12,000	588 (~ 69 min)

3 Task 1: Decoding

3.1 Method

I used `model.generate()` with three English prefixes: `en_a` (“Once upon a time, Stella Lou had a dream.”), `en_b` (“In a small village near the forest, there lived an old carpenter.”), `en_c` (“A scientist discovered a strange door in the laboratory.”). Six settings: greedy; $T = 0.3, 0.6, 1.0$; $T = 0.8+k50$; $T = 0.8+p90$. Max 120 new tokens, seed 42. I summarise each continuation with word count, TTR, and repeated bigram / trigram rates (rep_2, rep_3).

3.2 Results

Table 3 averages over the three prompts. Mean TTR rises from 0.607 (greedy) to 0.676 ($T = 1.0$) while rep_2 falls from 0.109 to 0.029 (Fig. 1). The trend is not monotonic in temperature: at $T = 0.3$ and $T = 0.6$ the model loops more than greedy (rep_2 0.124 and 0.116), because moderate randomness can lock the sampler onto a repeated phrase. So greedy is *not* the most repetitive setting in aggregate ($rep_2/rep_3 = 0.109/0.036$); its worst single case is still `en_b`, where it repeats “carpenter’s carpenter’s car” ($rep_2=0.19$). At $T = 0.8$, top- $k=50$ and top- $p=0.90$ produced identical outputs on all three prompts under the shared seed—we read this tentatively as the two truncation rules keeping overlapping candidate sets on a peaked distribution, though we only observed three prompts.

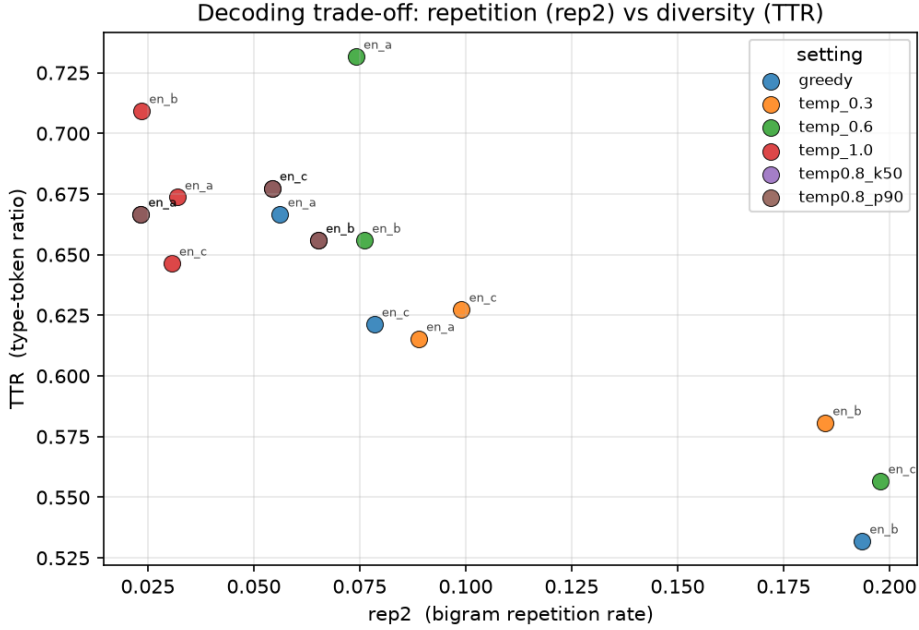


Figure 1: Task 1 diversity–repetition trade-off: rep_2 vs. TTR.

Table 3: Task 1: aggregate lexical statistics (mean over 3 prompts).

Setting	Words	TTR	rep_2	rep_3
greedy	95.7	0.607	0.109	0.036
$T = 0.3$	95.3	0.608	0.124	0.040
$T = 0.6$	90.7	0.648	0.116	0.056
$T = 1.0$	93.3	0.676	0.029	0.000
$T = 0.8+k50$	91.0	0.667	0.048	0.007
$T = 0.8+p90$	91.0	0.667	0.048	0.007

4 Task 2: Perplexity Evaluation

Using `compute_ppl`, I evaluated the base model on the three 1,000-doc test sets (Table 4). English PPL is 4.14 (median 3.05) but Chinese is 70,003.65 and Portuguese 25,046.78. Distribution shift in surface forms and subword segmentation explains the gap: Chinese is tokenized into character-level or OOV pieces the English-trained model assigns near-zero probability. A qualitative check confirms it: given the Chinese prefix “从前有一只小兔子乖乖”, the model ignores the language cue and continues in English (“...were playing in the garden. Suddenly, he noticed a big, yellow butterfly.”). The base model has almost no cross-lingual generalisation, motivating Task 3.

Table 4: Task 2: base-model PPL on held-out test sets.

Test set	Mean	Median	Min	Max
English	4.14	3.05	1.63	53.37
Chinese	70,003.65	70,374.41	23,577.15	108,491.70
Portuguese	25,046.78	23,192.41	5,018.26	97,724.66

5 Tasks 3 and 4: Continual Pre-training

5.1 Training dynamics

Fig. 2 shows training/validation losses: all curves decrease smoothly and validation tracks training closely with no strong overfitting even after 8 epochs. Run A (10^{-4} , 8 ep) converges lowest on Chinese validation (≈ 1.28 at step 1,800); Run B with half the LR declines more slowly and stays a little higher (1.43 vs. 1.28 at step 1,800); the 2-epoch runs C and D stop while loss is still falling.

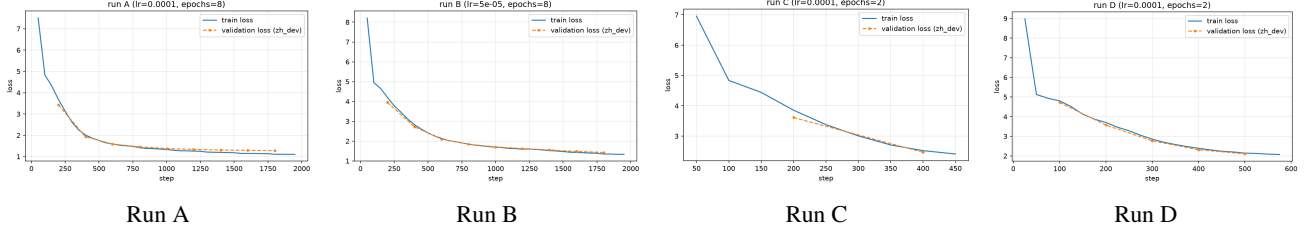


Figure 2: Training (solid) and validation (dashed) loss vs. step.

5.2 Perplexity after continual pre-training

Table 5 compares base and runs A–D (Fig. 3). Chinese improves by orders of magnitude: Run A drops zh_test PPL from 70,003.65 to 3.47, even the shortest run C reaches 12.93. Forgetting depends on the run: long training at high LR (Run A) pushes English PPL from 4.14 to 24.24, whereas halving the LR (Run B) keeps English at 8.14 while reaching Chinese 4.99; Portuguese, unseen during training, degrades far more in A (1.8×10^6) than in B/C/D. Fewer epochs trade Chinese gains for English retention (Run C leaves English almost intact at 4.94), and adding data balances the two: Run D (same settings as C, 12,000 docs) improves Chinese 12.93 $\rightarrow$ 9.33 while English stays at 5.74. Repeating Run C with seed 123 (seed_study.json) gives zh 13.91 / en 4.77 vs. 12.93 / 4.94 for seed 42; the spread is small for English (4.77–4.94) but not negligible for Chinese (12.93–13.91), so single-seed differences of order 1 in Chinese PPL should be read with caution.

Table 5: Mean test PPL of the base model and four runs.

Model	zh_test	en_test	pt_test
base	70,003.65	4.14	25,046.78
A (10^{-4} , 8 ep)	3.47	24.24	1,844,312.31
B (5×10^{-5} , 8 ep)	4.99	8.14	167,925.07
C (10^{-4} , 2 ep)	12.93	4.94	154,450.56
D (10^{-4} , 2 ep, +2k)	9.33	5.74	219,562.30

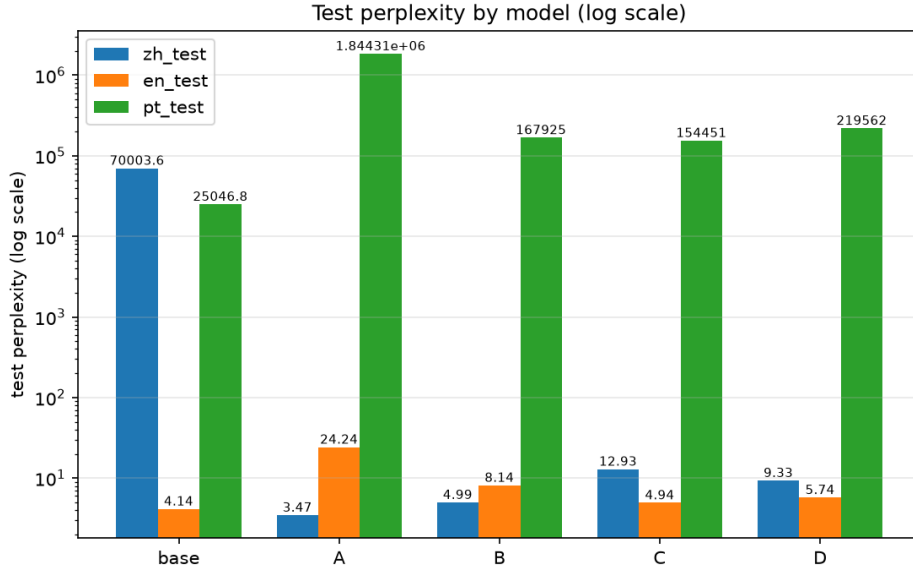


Figure 3: Mean test PPL on a log scale. For fine-tuned runs, the Portuguese bar reads as “near-zero probability”, not a precise value.

5.3 Chinese generation, discussion and limitations

After continual pre-training the model can begin a Chinese story in Chinese. Given the prompt “从前，有一只叫做汤姆的猫。汤姆和他的朋友们一起玩。”，Run D continues on topic with broadly correct morphology, but still shows repetition, occasional ungrammatical fragments, and tokenizer artifacts (U+FFFD). Full story quality would need a larger model, more data and more training; the large PPL gain reflects reasonable probabilities on in-domain Chinese sequences

rather than fluent generation.

Limitations. Every run uses a single seed (the two-seed check above is the only spread estimate) and PPL is reported on 1,000 docs without CIs; Portuguese PPL after training reaches 10^6 and must be read qualitatively. Run D’s Chinese comes from automatic NLLB-200 translation whose residual noise bounds generation quality, and the 2,000 extra stories were not deduplicated against the original corpus (both derive from TinyStories); overlap would slightly favour Run D. No further LR/epoch/data sweeps were run. A useful next step is to search the data-size $\times$ epochs trade-off more finely and evaluate with human or reference-based story metrics.

6 Use of Generative AI

I acknowledge the use of *GitHub Copilot* (<https://github.com/features/copilot>) and *ChatGPT* (<https://chatgpt.com>) to help write and debug parts of the experiment code, to summarise metrics, and to draft and revise this report. All experimental designs, runs, measurements, and interpretations are my own, and I reviewed and edited any AI-generated text before presenting it as my work.

References

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